

Vector product

Key Points

The vector/cross product is

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

The vector product is: anti-commutative and distributive so

$$\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$$

$$\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$$

It can also be written as

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \, \hat{\mathbf{n}}$$

Basic Skills

Q1. By using the formula find

a) $\begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$ b) $\begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ c) $\begin{pmatrix} 6 \\ 10 \\ 2 \end{pmatrix} \times \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$

Q2. Evaluate $\begin{vmatrix} 3 & -2 & -1 \\ 4 & 3 & 5 \\ -3 & 1 & -2 \end{vmatrix}$

Q3. For the vectors $\mathbf{a} = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$

a) Write an appropriate matrix determinant in an effort to calculate the vector product

b) Hence compute the vector product

Competency Skills

Q4. For $\mathbf{a} = \begin{pmatrix} -2 \\ 1 \\ -3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix}$

a) Show that they vectors are parallel

b) Compute $\mathbf{a} \times \mathbf{b}$

c) State the rules about scalar and vector products for parallel and perpendicular vectors

Q5. For $\mathbf{a} = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}$ compute $\mathbf{a} \times \mathbf{b}$ and hence confirm that $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 0 = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b}$ explaining what this means about the vector product

Q6. Given that the angle between two unit vectors is 30° and the direction is parallel to $3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ find there vector product

Advanced Skills

Q7. Prove that the vector product is anti-commutative

Q8. Prove that $|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2$

Q9. Explain why it is not possible to compute the vector product in 2D

Q10. Two in-planar vectors are $\mathbf{a} = \begin{pmatrix} 5 \\ 5 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ -2 \\ 0 \end{pmatrix}$

a) Find a vector of unit length perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

b) Hence state an equation for the plane going through the origin with points spanned by the vectors $\mathbf{a}$ and $\mathbf{b}$

Extension

Q11. Deduce the equation for a plane given two non-parallel in-planar vectors

Hint: Consider how the normal is related to any point on the plane